

Confidence Estimation Using Unlabeled Data

ICLR 2023

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Background

▪ Confidence / Uncertainty Estimation

- 특정 도메인에서는 한 번의 잘못된 예측이 큰 위험을 초래할 수 있다.
 - e.g., 자율 주행, 의료, ...
- 신뢰할 수 있는 모델을 얻기 위해, 모르는 것을 "모른다"고 말할 수 있게끔 모델을 학습시키자.
 - Anomaly / out-of-distribution (OOD)
 - Covariate shift
 - Active learning

▪ Problem setting: Multi-class classification

- $y_i \in \mathcal{Y} = \{1, \dots, K\}$, $f(x, y; W) : \mathcal{X} \times \mathcal{Y} \rightarrow [0, 1]$.
- $\hat{y}_i = \arg \max_{y \in \mathcal{Y}} f(x, y; W)$.

Background

■ Confidence Estimation

- **Max softmax as a confidence:** $\kappa_i = \max_{y \in \mathcal{Y}} f(x, y; W)$.
 - 일반적으로 confidence로써 활용되는 값.
 - Deep learning 모델은 overconfident한 경향이 있다.
- **Confidence ordinal ranking**
 - 더 확실한 데이터에 대해 더 높은 confidence를 주게끔 하는 작업.

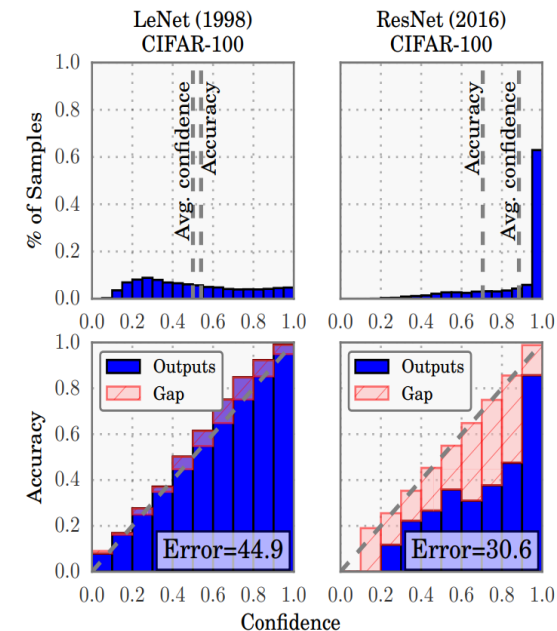


Fig 1. Reliability diagram. ResNet 모델은 높은 confidence에 비해 비교적 낮은 accuracy를 보인다.

Background

▪ The Area Under the Risk-Coverage Curve (AURC)

- Confidence를 통해 잘못 예측할 가능성이 높은 데이터를 가려낼 수 있는가?
- **Empirical selective risk** ($\hat{r}$): confidence가 높은 데이터들 중 잘못 예측한 데이터의 비율.
 - Given a dataset with m samples $\mathcal{D}_m = (X_m, Y_m)$ and a confidence estimator κ ,
 - **Empirical selective risk** $\hat{r}(f, g_\theta | \mathcal{D}_m) = \frac{\frac{1}{m} \sum_{i=1}^m l(\hat{y}_i, y_i) g_\theta(x_i)}{\frac{1}{m} \sum_{i=1}^m g_\theta(x_i)}$.
 - $l(\hat{y}_i, y_i) = \mathbf{1}\{\hat{y}_i \neq y_i\}$, $g_\theta(x_i) = \mathbf{1}\{\kappa > \theta\}$.
- **AUPR**: confidence threshold θ 에 따른 empirical selective risk를 aggregate한 값.
 - $AURC(f, \kappa | \mathcal{D}_m) = \frac{1}{m} \sum_{\theta \in \Theta} \hat{r}(f, g_\theta | \mathcal{D}_m)$.
- **E-AUPR**: AUPR의 normalized version.
 - $E-AURC(f, \kappa | \mathcal{D}_m) = AURC(f, \kappa | \mathcal{D}_m) - \min_{\kappa} AURC(f, \kappa | \mathcal{D}_m)$.

Method

▪ Confidence-Aware Semi-supervised Learning

- n labeled data and p unlabeled data.
 - $\mathcal{D} = (X, Y, U)$, $X = \{x_1, \dots, x_n\}$, $Y = \{y_1, \dots, y_n\}$, $U = \{x_{n+1}, \dots, x_{n+p}\}$.
 - U 는 confidence estimator 학습에 활용.

▪ Training **Consistency** as a Surrogate of a Confidence Estimator

- **Consistency:** Training 동안 모델이 동일하게 예측한 비율
 - $c_i = \frac{1}{T-1} \sum_{t=1}^{T-1} \mathbf{1}\{\hat{y}_i^t = \hat{y}_i^{t+1}\}$, t : epoch.
 - Unlabeled data에도 적용 가능한 기법.

Method

■ Motivation: Qualitative & Quantitative Analysis

- **Max softmax**: overconfident in the data near the decision boundary.
- **Correctness**¹: a known good proxy of confidence **in fully-supervised setting**.
- **Consistency**: seem to be well correlated with confidence and comparable to **Correctness**

$$c_i = \frac{1}{T} \sum_{t=1}^T \mathbf{1}\{y_i = \hat{y}_i^t\}$$

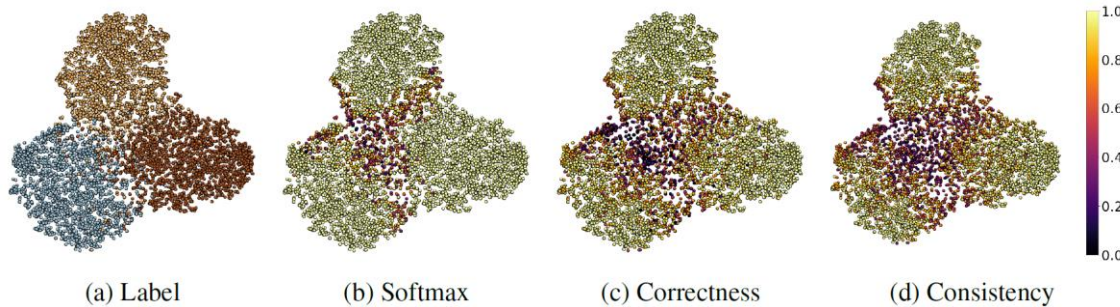


Fig 2. Feature representation of 6000 unlabeled CIFAR10 data. Since the model is trained on 20% labeled data, it is expected that lower confidence data are closer to the decision boundary.

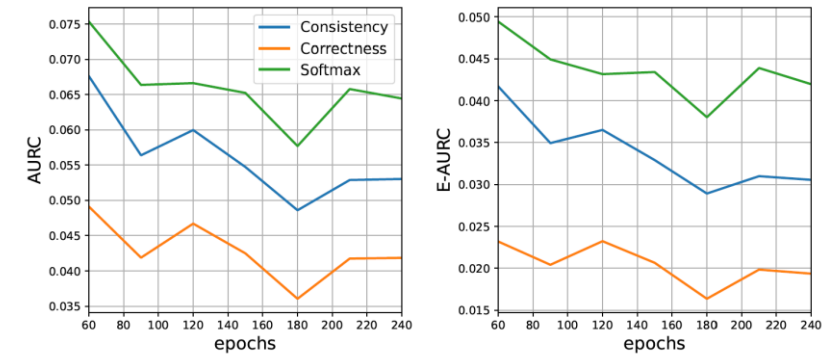


Fig 3. Performance of consistency as a confidence estimator. The model is trained with 20% of CIFAR10 labeled data and evaluated on the remaining unlabeled data.

Method

▪ Training **Consistency** as a Surrogate of Confidence

- **Issue:** test할 때에는 consistency를 측정할 수 없다.
- **Consistency ranking loss:** max softmax에 의한 rank가 consistency에 의한 rank를 따라하도록 학습.

$$- \mathcal{L}_{cons}(f) = \sum_{s=1}^{n+p} \sum_{i=1, c_i < c_s}^{n+p} \max(0, (c_s - c_i) - (\kappa_s - \kappa_i)), \quad \kappa_i = \max_{y \in \mathcal{Y}} f(x, y; W).$$

(1) Confidence ranking이 잘못되었을 때, 그리고

(2) Consistency gap이 confidence gap보다 클 때 loss를 준다.

- **Overall loss:** $\mathcal{L} = \mathcal{L}_{CE} + \lambda_1 \mathcal{L}_{corr} + \lambda_2 \mathcal{L}_{cons}$

– Correctness도 함께 confidence estimator 학습에 활용.

– Correctness ranking loss $\mathcal{L}_{corr}(f) = \sum_{i=1}^n \sum_{j \neq i} \max(0, -g(c_i, c_j)(\kappa_i - \kappa_j) + |c_i - c_j|), \quad g(c_i, c_j) = \begin{cases} 1, & c_i > c_j \\ 0, & c_i = c_j \\ -1, & \text{otherwise} \end{cases}$

Experiments

▪ Questions

- Consistency ranking loss의 효과.
- 본 방법론을 통한 confidence-aware semi-supervised learning의 효과.
 - 학습 성능: Accuracy (classification) / mIOU (segmentation).
 - Confidence 성능: AURC, E-AURC

▪ Benchmark

- Classification: CIFAR-10/100.
- Medical image segmentation: ISIC2017.

Experiments

▪ Experiments 1: Consistency ranking loss의 효과

- $\mathcal{L}_{cons}$ 로 학습된 confidence estimator가 training consistency와 비슷한 성능을 보임을 확인.

Table 1: The quality of optimization on consistency ranking loss. Softmax (ours): the maximum softmax output trained with consistency ranking loss. TC (ours): training consistency. Diff. (ours): the evaluation metric difference between our maximum softmax output and training consistency. Softmax, TC, Diff. are corresponding results without consistency ranking loss. All values $\times 10^2$.

Dataset (labeled size)	Method	E-AURC↓	AURC↓	Method	E-AURC↓	AURC↓
CIFAR10 (10000)	Softmax	29.07	40.57	Softmax (ours)	19.35	32.57
	TC	17.63	29.13	TC (ours)	17.17	30.39
	Diff.	11.44	11.44	Diff. (ours)	2.17	2.17

←→
w/o consistency ranking loss

←→
w/ consistency ranking loss

Experiments

▪ Experiments 2: image classification에서 baseline들과의 비교.

- Baseline들에 비해 더 나은 classification 및 calibration 성능을 보였다.
- Labeled data가 적을수록 그 차이가 두드러졌다.
 - Unlabeled data를 활용함으로써 더 나은 confidence estimator를 학습할 수 있었다.

Labeled size	Method	Acc↑	AURC↓	E-AURC↓	Acc↑	AURC↓	E-AURC↓
CIFAR10				CIFAR100			
2500	Softmax	0.717±0.008	109.78±6.69	65.32±3.92	0.292±0.006	506.70±4.52	159.23±6.38
	AES	0.713±0.002	108.94±0.87	63.34±1.02	0.289±0.009	509.38±12.07	157.50±3.01
	Mcdrop	0.700±0.008	122.45±6.37	72.42±3.89	0.269±0.005	542.10±7.75	165.06±2.00
	Aleatoric+MC	0.704±0.022	119.76±15.06	70.66±7.69	0.269±0.005	542.86±10.72	165.56±3.95
	CRL	0.718±0.003	105.92±2.27	61.95±1.27	0.287±0.004	507.43±5.96	152.82±2.28
	Ours	0.755±0.004	81.50±2.70	48.72±1.75	0.365±0.004	399.49±6.23	133.15±3.78
5000	Softmax	0.795±0.002	60.54±1.45	38.02±1.14	0.425±0.005	344.09±5.31	132.95±0.95
	AES	0.799±0.002	51.68±1.04	29.99±0.80	0.419±0.005	335.02±4.86	118.75±0.87
	Mcdrop	0.808±0.003	52.94±1.07	33.25±0.47	0.394±0.004	379.60±4.24	141.06±2.55
	Aleatoric+MC	0.810±0.001	50.96±1.47	31.73±1.36	0.394±0.002	378.72±3.47	140.54±1.48
	CRL	0.807±0.002	51.76±1.91	31.82±1.55	0.437±0.003	322.94±3.17	122.50±0.77
	Ours	0.818±0.002	44.43±1.03	26.88±0.66	0.482±0.005	266.08±5.87	100.22±2.40
10000	Softmax	0.849±0.001	37.46±1.09	25.44±0.93	0.546±0.004	214.17±2.81	90.78±0.59
	AES	0.855±0.004	30.75±0.89	19.71±0.57	0.546±0.002	209.77±3.62	86.38±2.24
	Mcdrop	0.865±0.004	28.58±1.08	19.09±0.58	0.523±0.004	238.32±5.79	100.70±3.53
	Aleatoric+MC	0.865±0.001	28.64±0.70	19.22±0.65	0.521±0.006	240.02±6.93	100.94±3.05
	CRL	0.856±0.001	31.28±0.32	20.51±0.16	0.563±0.005	196.11±2.35	82.976±0.95
	Ours	0.860±0.002	28.39±0.54	18.19±0.36	0.590±0.003	168.57±2.39	70.26±0.89

Experiments

- Experiments 3: medical image segmentation에서 baseline들과의 비교.
 - Segmentation 성능은 큰 impact가 없었으나, confidence estimation 성능에서 큰 차이를 보였다.
 - Unlabeled data를 활용한다는 점으로 인해, 다른 baseline들에 비해 labeled size에 robust한 결과를 보였다.

Dataset (labeled size)	Method	mIOU↑	AURC↓	E-AURC↓
ISIC2017				
125	Softmax	0.801±0.005	34.36±7.76	31.33±7.92
	AES	0.802±0.006	21.12±1.10	18.07±1.01
	Mcdrop	0.801±0.005	30.23±3.81	27.19±3.75
	Aleatoric+MC	0.802±0.001	27.66±1.87	24.62±1.93
	CRL	0.810±0.004	22.14±2.82	19.33±2.81
	Ours	0.812±0.007	14.11±1.06	11.23±0.86
250	Softmax	0.819±0.002	28.91±4.70	26.45±4.72
	AES	0.819±0.002	18.08±1.38	15.60±1.43
	Mcdrop	0.819±0.008	25.38±4.76	22.89±4.57
	Aleatoric+MC	0.817±0.005	25.96±3.76	23.37±3.80
	CRL	0.818±0.002	16.43±0.94	13.90±0.90
	Ours	0.817±0.007	12.79±0.79	10.21±0.62
500	Softmax	0.822±0.003	23.45±0.93	21.09±0.90
	AES	0.823±0.006	15.74±1.18	13.39±1.13
	Mcdrop	0.821±0.003	21.21±2.57	18.83±2.55
	Aleatoric+MC	0.821±0.003	21.32±2.18	18.96±2.12
	CRL	0.822±0.006	14.04±1.03	11.64±1.00
	Ours	0.823±0.004	11.85±1.17	9.42±1.13

Analysis

▪ Theoretical Guarantee of Consistency Ranking Loss

Theorem 1. *Given dataset $\mathcal{D}_m = (X_m, Y_m)$ and training consistency C_m :*

$$|AURC(f, c|\mathcal{D}_m) - AURC(f, \kappa|\mathcal{D}_m)| \leq \frac{1}{m \min D_c} \mathcal{L}_{cons}(f, X_m; C_m)$$

- Training consistency의 성능과 학습된 confidence estimator의 **성능 gap의 upper bound가 $\mathcal{L}_{cons}$ 에 의해 결정.**
- 즉, $\mathcal{L}_{cons}$ 를 줄이는 것이 성능 gap을 줄이는 것을 보장한다.

Conclusion

▪ Contributions

- Confidence-aware semi-supervised learning 프레임워크 제안.
- Training Consistency라는 새로운 confidence 기법 제시.
- Consistency ranking loss의 이론적 유효성 검증.
- 본 방법론이 baseline에 비해 우월한 성능을 보이는 것을 실험을 통해 입증.
 - 특히 labeled data가 적을 때 더 효과적이라는 것을 밝힘.

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